

Vision-Based Target Selection and Path Clearance Estimation for a Quadraped Robot Using Stereo Depth Sensing

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Abstract—We extend quadruped locomotion with real-time perception using the OAK-D Lite stereo camera on the Stanford Pupper. A “Perception Gate” combines on-device YOLO inference with stereo depth to verify a commanded path is collision-free before the PPO controller from Lab 5 executes it. A secondary objective uses a rigidly mounted single-servo gripper aimed by whole-body motion. Evaluation covers distance-estimation accuracy, path-classification accuracy, and navigation success rate across 20 trials.

I. INTRODUCTION

Quadrupedal robots require environmental awareness to navigate safely. The OAK-D Lite provides an onboard VPU for neural inference and stereo depth mapping simultaneously [1]. This project extends Lab 5, whose PPO policy follows velocity commands but cannot perceive obstacles or select targets. We propose a *Perception Gate* using a YOLO spatial detector to identify targets, compute their 3D position, and apply a binary path-clearance check before issuing velocity commands. A secondary objective tests a rigidly mounted single-servo gripper aimed by repositioning the robot’s body—an approximation of recent passive-gripper work [4] (Fig. 1).

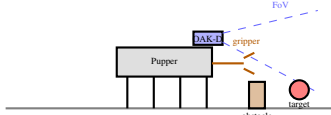


Fig. 1. Pupper with OAK-D Lite and rigidly mounted single-servo gripper, detecting a target while evaluating corridor clearance.

II. TECHNICAL APPROACH

The architecture integrates high-level perception with the low-level PPO controller (Fig. 2). Sensor inputs: RGB stream I_{rgb} , stereo depth D_z , and IMU $(\mathbf{a}, \boldsymbol{\omega})$. Estimated states: target centroid $\mathbf{P}_{\text{obj}} = [x, y, z]^T$ in the camera frame and binary clearance flag $C \in \{0, 1\}$.

A. Path Clearance Logic

A 3D *safety corridor*—a rectangular prism of width w and height h —is defined between the robot and the target centroid. The clearance flag samples the stereo depth map inside the corridor:

$$C = \begin{cases} 1, & \text{if } \forall \mathbf{p} \in \text{Corridor}, z(\mathbf{p}) > d_{\text{safe}} \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

If $C = 0$, $\mathbf{v}_{\text{cmd}}$ is zeroed. We use YOLOv8n exported to the OAK-D’s .blob format, running on-device via the YoloSpatialDetectionNetwork node which returns (x, y, z) per detection [1]. We build on the PPO framework of Schulman et al. [2], task-space mapping from Lab 4, and the depth-based

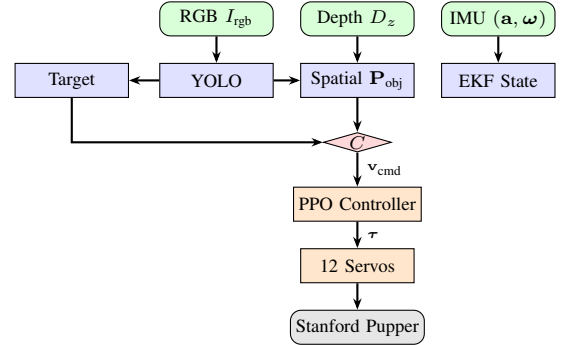


Fig. 2. System block diagram. Green = sensors, blue = perception, red = decision, orange = control. IMU/encoder data close the proprioceptive loop via the EKF. path-validity approach of Chen et al. [3], extending the course’s blind controllers (Labs 3, 5) with closed-loop visual feedback.

III. PERFORMANCE METRICS

Evaluation uses $N = 20$ trials. Ground-truth distances d_i^{true} are obtained from premeasured floor markers (± 1 cm). We report distance error E_d , clear/blocked classification accuracy A_{path} , and navigation success rate SR :

$$E_d = \frac{1}{N} \sum_{i=1}^N |d_i^{\text{est}} - d_i^{\text{true}}|, \quad A_{\text{path}} = \frac{N_{\text{correct}}}{N_{\text{total}}}, \quad SR = \frac{N_{\text{success}}}{N} \times 100\% \quad (2)$$

A trial is successful if the robot reaches within 0.25 m of the target without collision when $C = 1$, or halts before the obstacle when $C = 0$.

IV. EXPERIMENTAL DESIGN

Simulation (MuJoCo): Using the Lab 5 environment, we inject virtual target coordinates and synthetic corridor-occupancy values to verify that zero-velocity commands produce stable stops without oscillation (10 trials per condition). **Physical Trials:** The OAK-D Lite is mounted on the Pupper. A target ball is placed at $\{1, 2, 3\}$ m; in half of 20 trials, a static cardboard box ($30 \times 30 \times 40$ cm) is placed in the corridor. Trials are video-logged. If time permits, we additionally test fixed-gripper grasping: once the robot reaches within 0.15 m of the target, the single servo closes (10 attempts).

REFERENCES

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